higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N4

29 MARCH 2019

This marking guideline consists of 9 pages.

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QUESTION 1

1.1
$$|D| = \begin{vmatrix} 8 & 2 \\ -4 & 3 \end{vmatrix} \checkmark$$
$$= (8 \times 3) - (-4 \times 2)$$
$$= 32$$
$$|D_x| = \begin{vmatrix} 4 & 2 \checkmark \\ 4 & 3 \end{vmatrix}$$
$$= (4 \times 3) - (4 \times 2)$$
$$= 4$$
$$|D_y| = \begin{vmatrix} 8 & 4 \\ -4 \checkmark & 4 \end{vmatrix}$$
$$= (8 \times 4) - (-4 \times 4)$$
$$= 48$$
$$x = \frac{|D_x|}{|D|} = \frac{4}{32} = 0{,}125$$
$$y = \frac{|D_y| \checkmark}{|D|} = \frac{48}{32} = 1{,}5 \checkmark$$
(6)

1.2 1.2.1

Minor of 5: ✓

Minor of 5: $\begin{vmatrix} 2 & -2 \\ 1 & 1 \end{vmatrix}$
$= 4$ ✓ (1)

1.2.2

Co-factor of 4:

$(-)\begin{vmatrix} 3 & 6 \\ 1 & -1 \end{vmatrix}$ ✓
$= +3$ ✓ (2)

1.3
$$V_1 = 49{,}24 + j15{,}628$$
$$V_2 = 52{,}5 + j90{,}933 \checkmark$$
$$V_3 = 57{,}851 - j68{,}9444 \checkmark$$
$$V_T = 49{,}24 + j15{,}628 + 52{,}5 + j90{,}933 + 57{,}851 - j68{,}944 \checkmark$$
$$V_T = 159{,}591 + j37{,}617, V_T = 163{,}964 \underline{|13{,}263} \checkmark$$
✓ (6)

1.4

$$z = \frac{24\angle\left[\frac{\pi}{6} + \frac{\pi}{3}\right.}{6\angle\frac{\pi}{4}} \checkmark$$

$$z = 4\angle\left(\frac{\pi}{6} + \frac{\pi}{3} - \frac{\pi}{4}\right) \checkmark$$

$$z = 4\angle\frac{\pi}{4}$$

$$z = 4\left[\cos\frac{\pi}{4} + j\sin\frac{\pi}{4}\right] \checkmark$$

$$z = 2{,}828 + j\sin 2{,}828 \checkmark \checkmark$$

(5)
[20]

QUESTION 2

2.1

$$u = x^4, V = \tan 4x$$

$$\frac{du}{dx} = 4x3, \frac{dv}{dx} = 4\sec^2 4x \checkmark$$

$$\frac{dy}{dx} = 4x^3(\tan 4x) + 4\sec^2 4x.x^4 \checkmark$$

$$= 4x^3 \tan 4x + 4x^4 \sec^2 4x \checkmark$$

(3)

2.2

$$\frac{dy}{dx} = \frac{\cos x}{2}$$

$$\frac{x}{2} = \arccos x0$$

$$\frac{x}{2} = \pm\frac{\pi}{2}$$

$$x_1 = \pi, x_2 = -\pi$$

$$y_1 = \sin\frac{x}{2}$$

$$y_1 = 1 \therefore (-\pi,-1)$$

$$y_2 = \sin\left(\frac{\pi}{2}\right)$$

$$y_2 = -1 \therefore (\pi,1)$$

$$\frac{d^2y}{dx^2} = -\sin\frac{x}{2}$$

$$for \quad x = \pi \therefore \frac{d^2y}{dx^2} = -\sin\frac{\pi}{2} = -1 < 0 \qquad \max imum$$

$$\therefore (\pi,1)\max imum \quad TP_S$$

$$for \quad x = -\pi \therefore \frac{d^2y}{dx^2} = -\sin\left(\frac{-\pi}{2}\right) = 1 > 0 \qquad \min imum$$

$$\therefore (-\pi,1)\min imum \quad TP_S$$

(8)

2.3

$$\frac{dy}{dx} = 7(\pi^x)\ln\pi - \frac{-1}{3x\ln 3} - \frac{30}{x^6} + 36\sec^2 6x - \cos ec^2 2x$$

(5)

2.4

$$f(x) = \frac{1}{x+3}$$

(4)

$$f(x+h)=\frac{1}{x+h+3}$$

$$f(x+h)-f(x)=\frac{1}{x+h+3}-\frac{1}{x+3}$$

$$=\frac{x+3-x-h-3}{(x+h+3)(x+3)} \quad \checkmark$$

$$=\frac{-h}{(x+h+3)(x+3)} \quad \checkmark$$

$$\frac{f(x+h)-f(x)}{h}=\frac{-h}{(x+h+3)(x+3)}x\frac{1}{h}$$

$$=\frac{-1}{(x+h+3)(x+3)} \quad \checkmark$$

$$\lim_{x\to\infty}\left(\frac{f(x+h)-f(x)}{h}\right)=\frac{-1}{(x+0+3)(x+3)}$$

$$=\frac{-1}{(x+3)^2} \quad \checkmark$$

[20]

QUESTION 3

3.1 $\int\left(1+\sec^2 x\right)dx$

$x+\tan x+c$ (3)

✓ ✓ ✓

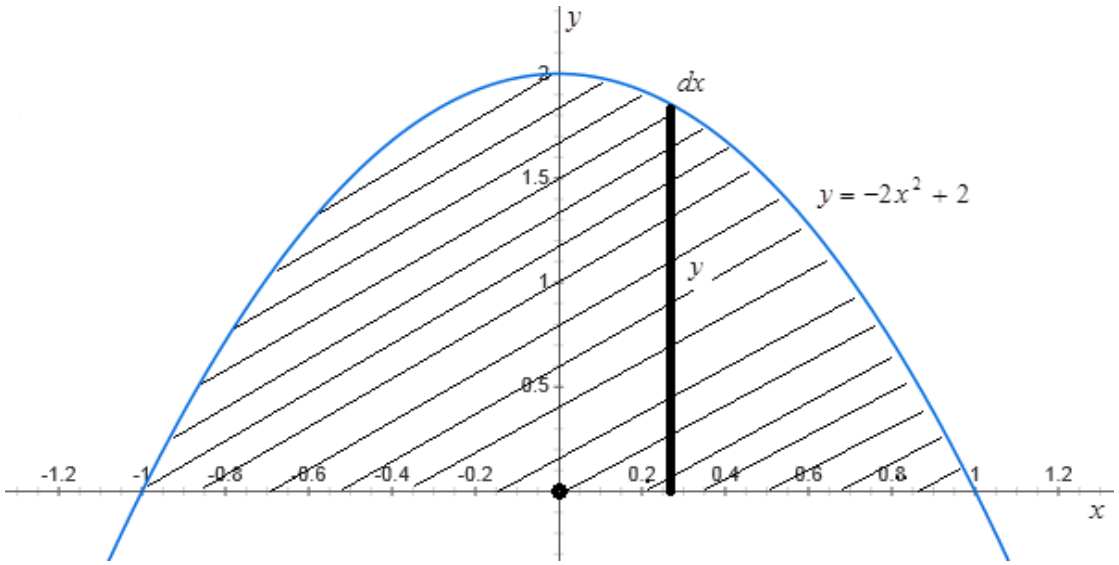
3.2 $=\left[\frac{\tan 2x}{2}\right]_0^{\frac{\pi}{8}}$ ✓

$=\left[\frac{\tan 2\left(\frac{\pi}{8}\right)}{2}-\frac{\tan 0}{2}\right]$ ✓

$=\frac{1}{2}$ (3)

✓

3.3 3.3.1



✓ Shape
✓ Representative
✓ Area shown (3)

3.3.2

$$Aox = \int_{-1}^{1}\left(-2x^2+2\right)de \quad \checkmark$$

$$= \left[\frac{-2x^3}{3}+2x\right]_{-1}^{1} \quad \checkmark$$

$$= [1{,}334]-[-1{,}334]$$

$$= 2{,}667u^2 \quad \checkmark$$

(4)

3.4 $$\frac{-2\pi\cos wx}{w} \checkmark + \frac{2.2^x}{\ln^2} \checkmark + 3x \checkmark + \frac{1}{x} \checkmark + \cos ecx \checkmark + \frac{ex^2}{2} \checkmark + c \checkmark$$ (7)

[20]

QUESTION 4

4.1 $$= \frac{4\sin A\cos A}{\sin^2 A} \checkmark$$

$$= \frac{4\cos A}{\sin A} \checkmark$$

$$= 4\cot A \checkmark$$ (3)

4.2 $7\cos^2 A - 3(1-\cos^2 A) + 1 + \cos A = 0$

$10\cos^2 A + \cos A - 2 = 0$

$$\cos A = \frac{-1 \pm \sqrt{(1)^2 - 4(10)(-2)}}{20} = \frac{-1 \pm \sqrt{81}}{20}$$

$$\cos A = \frac{-1 \pm 9}{20} \quad \checkmark$$

$$\cos A = -\frac{1}{2} \quad and/or \quad \cos A = \frac{2}{5} \quad \checkmark$$

$\alpha = 60°$

$A_1 = 180° - 60° = 120°$ $A_3 = 66{,}422°$

$A_2 = 180° + 60° = 240°$ $A_4 = 360° - 66{,}422°$ ✓ ✓

$A_4 = 293{,}578°$ ✓ ✓ (6)

4.3 $$\frac{2\cos^2 A - 1}{\frac{1}{2}.2\sin A\cos A} = L.H.S. \quad \checkmark$$

$$\frac{\cos 2A}{\sin A\cos A} = L.H.S. \quad \checkmark$$

$$R.H.S. = \frac{\cos A}{\sin A} - \frac{\sin A}{\cos A}$$

$$= \frac{\cos^2 A - \sin^2 A}{\sin A\cos A} \quad \checkmark$$

$$= \frac{\cos 2A}{\sin A\cos A}$$

$R.H.S. = L.H.S.$ ✓ (4)

4.4 $$\sin\frac{A}{2} = \pm\sqrt{\frac{1-\cos A}{2}} \quad \checkmark$$

$$= \pm\sqrt{\frac{1-\frac{-2}{5}}{2}} \quad \checkmark$$

$$\sin\frac{A}{2} = \pm\sqrt{\frac{3}{10}} \quad \checkmark \quad (3)$$

4.5 $\sin(-15°) = -\sin 15°$ ✓

$= -[\sin(60° - 45°)]$ ✓

$= -[\sin 60° \cos 45° - \cos 60° \sin 45°]$

$= -\left[\frac{\sqrt{3}}{2}.\frac{\sqrt{2}}{2} - \frac{1}{2}.\frac{\sqrt{2}}{2}\right]$ ✓

$= \frac{-\sqrt{6}+\sqrt{2}}{4}$ ✓ (4)

[20]

QUESTION 5

5.1 $x = 2\frac{1}{4} \therefore x - 2\frac{1}{4} = 0$

$4x - 9 = 0$ ✓

$x = -3\frac{1}{2} \therefore x + 3\frac{1}{2} = 0$

$2x + 7 = 0$

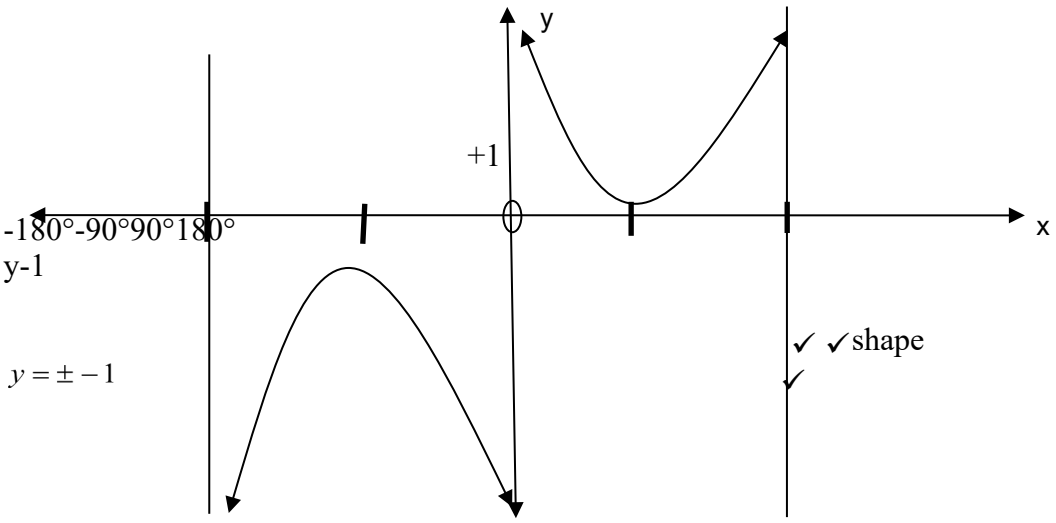
$(2x+7)(4x-9) = 0$ ✓

$8x^2 - 18x + 28x - 63 = 0$ ✓

$8x^2 + 10x - 63 = 0$ ✓

$\therefore y = 8x^2 + 10x - 63$ ✓ (6)

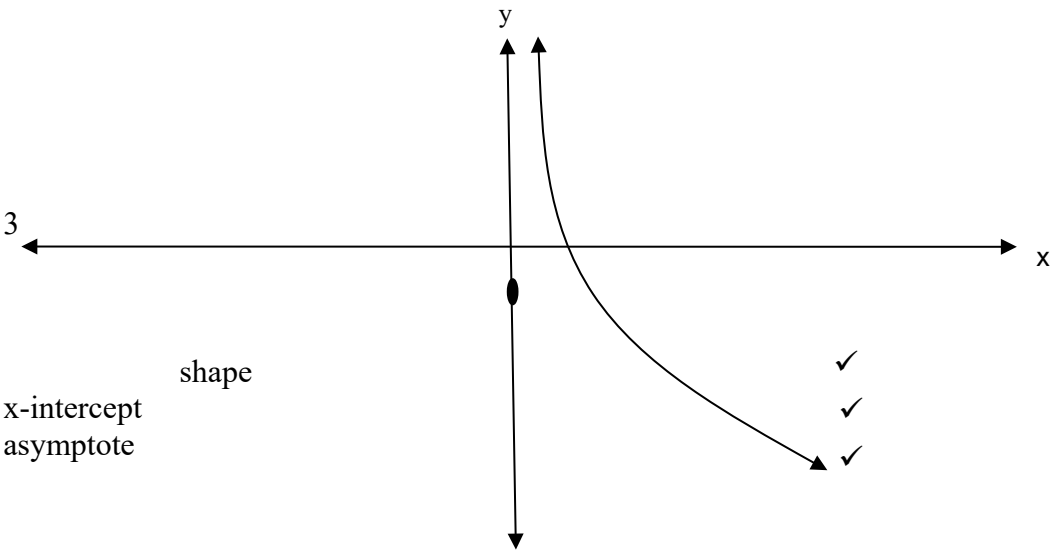
5.2 5.2.1 ✓



(3)

5.2.2 No/Discontinuous (1)

5.3



shape ✓
x-intercept ✓
asymptote ✓

(3)

5.4

$$3-\log\frac{P}{Q}=\frac{x}{b^y} \quad ✓$$

$$\log\frac{P}{Q}=3-\frac{x}{b^y} \quad ✓$$

$$\frac{P}{Q}=10^{\left(3-\frac{x}{b^y}\right)} \quad ✓$$

$$Q=\frac{P}{10^{\left(3-\frac{x}{b^y}\right)}} \quad ✓$$

(4)

5.5 $\left(3x^2-2y^4\right)\left(9x^4+6x^2y^4+4y^8\right)$ ✓ ✓ ✓

(3)

[20]

100

TOTAL:

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